

Sustainability Assessment in Craft Beer Production: Environmental, Economic, and Exergetic Analyses

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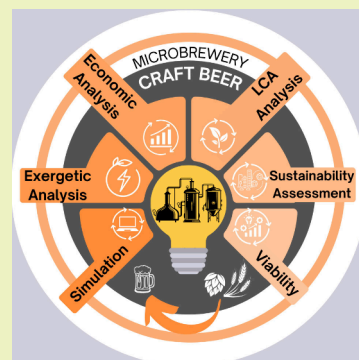
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ABSTRACT: In recent years, the craft microbrewery industry has grown notably, contributing to the global economy and increasing the demand for sustainable practices. This expansion underscores the need for focused research on the sustainability of this industry. This study evaluates craft beer production from environmental, economic, and exergetic perspectives. Life cycle assessment (LCA) results of craft beer production revealed raw materials (51%) and transportation (31%) as the main environmental impact contributors. A sensitivity analysis comparing raw material sourcing from Europe, Brazil, and Argentina revealed that sourcing from Brazil instead of Europe reduced impacts by up to 22%. The economic analysis confirmed the feasibility of craft beer production, with promising indicators including a net present value of \$US 49,422, a modified internal rate of return of 16.92%, and a benefit–cost ratio of 3.02. Additionally, exergetic analysis identified the first filtration as the greatest contributor to the overall exergy destruction (68%), highlighting opportunities for energy efficiency strategies. This comprehensive study underscores the importance of environmental, economic, and exergy analyses in achieving the sustainability of craft beer production. Furthermore, this study provides insights for process improvement through technological advancements and sustainable practices, supporting guidance for decision-making and fostering innovation in the craft beer industry.

KEYWORDS: life cycle assessment, sustainability, craft beer, economic analysis, exergy



INTRODUCTION

The craft beer industry has demonstrated remarkable growth and potential, as evidenced by an unprecedented number of new breweries established globally, now accounting for over 15% of the global beer market.¹ This growth is driven by consumer preferences for high-quality, locally brewed products with unique flavors and sustainable production practices.

In Europe, craft beer market has expanded significantly, with a compound annual growth rate of 13% between 2018 and 2022.² In Latin America, countries like Brazil, Mexico, and Colombia, the artisanal beer sector has undergone a significant transformation, driven by a cultural shift towards local identity valorization.^{3,4} Brazil, in particular leads the regional craft beer market, with a 40% growth rate, a per capita consumption of 69.1 L, and a projected compound annual growth rate of 14.39% from 2024 to 2032.⁵ Mexico has also showed strong craft beer trends, with 39% of consumers prioritizing locally sourced, sustainable brews that reflect global preferences for authenticity and environmental consciousness.⁵ In Colombia, the craft beer industry increased by 6.8% from 2019 to 2020, followed by a 3.1% growth from 2020 to 2021.⁶ However, this sector faced challenges during the COVID-19 pandemic, leading to a growth slowdown. Therefore, to drive future growth, the craft beer industry must focus on enhancing sustainability practices and improving production efficiency,

ensuring that the sector's expansion aligns with environmental and social responsibility.

Sustainability in craft beer production has become an increasingly relevant research area, as beer production requires intensive use of raw materials, energy, and water, leading to significant environmental impacts.^{7,8} The growing concern about climate change and the need for sustainable practices has made it imperative for industries, including breweries, to consider proactive steps in reducing their environmental impact.⁹ Life cycle assessment (LCA) methodology is a tool that provides a comprehensive analysis of environmental impacts from resource extraction to final product use.¹⁰ LCA is an analytical method used to evaluate the environmental effects of a product, process, or activity throughout its life cycle, known as “cradle to grave”.¹¹ However, LCA can also be applied from the “cradle to gate” approach, which is a partial approach that covers up to the production or manufacturing process. In addition to environmental considerations, econom-

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ic and exergy assessments are relevant for the craft beer industry to evaluate the cost-effectiveness and thermodynamic efficiency of production processes. Therefore, by integrating these three assessment tools, craft breweries can identify hotspots where inefficiencies and sustainability challenges arise. These tools facilitate the development and implementation of targeted strategies to enhance sustainability, optimize resource use, and drive innovation. This approach ensures that their operations are not only environmentally responsible but also economically viable and energy efficient.

This study aims to enhance energy use and environmental performance while strengthening the economic feasibility of craft beer production. Furthermore, by identifying key hotspots in the brewing process, this study proposes strategies to minimize environmental impact and enhance sustainability. Through LCA, economic, and exergy analyses, the study provides data-driven insights to support low-carbon emissions and resource-efficient practices.

This study deliberately employs a “cradle-to-gate” approach, excluding bottling, packaging, and distribution stages. While we acknowledge that packaging can contribute significantly to the overall environmental footprint of beer production (contributing 50–80% of emissions in some studies), our research focuses specifically on optimizing the brewing process itself. This methodological choice allows for a more detailed analysis of production inefficiencies that brewers can directly control within their facilities. The packaging stage, which often involves different suppliers and manufacturing processes, would benefit from a dedicated analysis that considers the variety of options available to craft brewers (bottles, cans, kegs) and their specific regional context.

METHODS

The methodological framework employed in this investigation encompassed the quantification of material inputs, energy consumption, and water utilization throughout the craft beer production lifecycle. However, the scope of the study explicitly excluded the assessment of environmental and economic externalities associated with waste generation and management.

Description of the Craft Beer Production. This study focuses on the production of craft IPA beer, chosen for its widespread popularity and high consumption rates.^{2,12} Figure 1 illustrates the craft beer production process investigated in this study, including various production steps and waste generation. The diagram illustrates the sequential stages from raw materials (barley-malt, hop, and yeast) through milling, mashing, wort separation, boiling, fermenting, to the final use and end of life.

Malt, the primary raw material, is obtained from barley through the malting process. This process involves three main steps: (i) steeping, where barley grains are soaked; (ii) germination, where the grain is allowed to germinate and sprout; and (iii) kilning, a controlled heating process that halts enzymatic activity while imparting specific flavors and colors to the malt.¹³

The craft beer production process begins with the meticulous selection and preparation of high-quality raw materials. First, the water's chemical composition is precisely adjusted to produce the desired flavor. Second, malt is milled to break the grains and pulverize them into flour, forming the basis for converting starches into fermentable sugars during mashing. During mashing, the milled malted barley is mixed with pretreated hot water through infusion to promote the enzymatic transformation of starches to fermentable sugars. This process requires precise temperature control, enzymes, water-to-malt ratio, time, and pH to obtain a sweet wort. The wort separation stage includes sparging and filtration. Sparging uses hot water to wash additional sugars from the malt husk and solids,

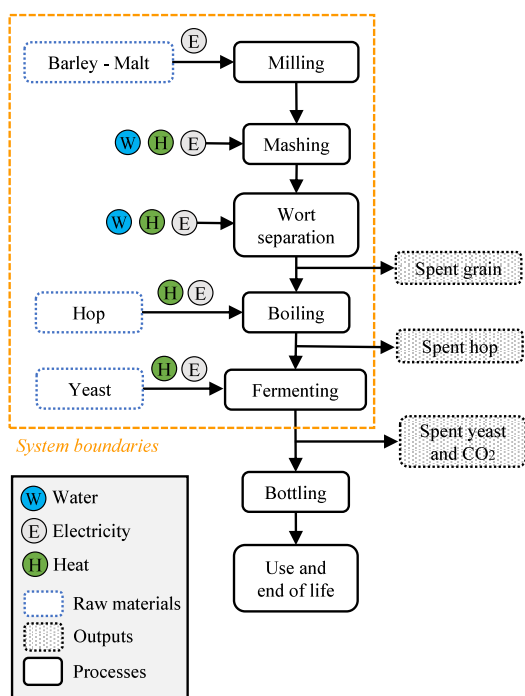


Figure 1. Process flowchart for craft beer production with system boundaries and resource inputs.

improving the sugar extraction process. Then, the wort is filtered to remove the spent grain.

Hops, added during the boiling stage, provide bitterness, flavor, and aroma. The boiling stage promotes flavor extraction and concentrates the wort through water evaporation and sterilizes it.¹⁴ Next, the wort is rapidly cooled in a heat exchange process before entering fermentation. Yeast, typically a *Saccharomyces cerevisiae* strain, is added to ferment the sugars into ethanol and carbon dioxide. Finally, the primary fermentation takes around 7 days, followed by polishing, clarification, carbonation, and packaging. The dry hopping technique, commonly used in IPAs to enhance aroma without increasing bitterness, and any secondary fermentation were not included in this study. This study is based on operational data from a commercial brewing facility in Bogotá, Colombia, where the entire process takes approximately 10 days. This timeframe aligns with literature values, which report fermentation durations for ALE beers ranging from 2 to 14 days.^{13,15}

Process Simulation. In this study, Aspen Plus software was employed to simulate the craft beer production process due to its extensive capabilities in process modeling, thermodynamic calculations, and industrial application success.^{16,17} This software effectively represents unit operations in craft beer production, such as milling, mashing, wort separation, boiling, and fermentation while enabling physical exergy calculations.

The Aspen Plus flowsheet of the craft beer production is shown in Supporting Information Figure S-1. This figure outlines the craft beer production process, with black dashed lines marking system boundaries for raw materials, wastes, and final products. Brewing stages, including milling, mashing, wort separation, boiling, and fermentation, are represented as subsystems (blue dashed lines) for a simplified overview. Key equipment, such as mills, vessels, heat exchangers, and filters, is depicted, with detailed specifications available in Supporting Information Table S-1.

To ensure the simulation accurately represents real-world brewing conditions, the model was developed using operational data obtained from a craft brewery located in Bogotá, Colombia, which operates with a batch capacity of 300.8 L.¹⁸ The thermodynamic model was selected based on a comprehensive evaluation of properties relevant to the system's nature, mixture composition, operational temperature and pressure, and model parameter availability. The NRTL

Table 1. Inventory Data of Craft Beer Production (1 L)

Flow	Amount	Unit	Ecoinvent dataset
Malt	0.176	kg	Dataset adapted from Supporting Information Table S-3
Hop	0.04	kg	Dataset adapted from Supporting Information Table S-4
Yeast	0.0004	kg	Dataset adapted from Supporting Information Table S-5
Water	1.298	kg	Market for tap water tap water APOS, S–CO
Electricity	0.489	kWh	Market for electricity, medium voltage electricity, medium voltage APOS, S–CO
Natural gas	0.049	m ³	Market for natural gas, low pressure natural gas, low pressure APOS, S–RoW
Transport, lorry	632.06	kg-km	Market for transport, freight, lorry, unspecified transport, freight, lorry, unspecified APOS, S–RoW
Transport, ship	47674.11	kg-km	Market for transport, freight, sea, container ship transport, freight, sea, container ship APOS, S–GLO

thermodynamic model was selected for its accuracy in representing the polar behavior of mixtures such as sugars, water, and alcohol.^{19,20}

Process modeling involved defining equipment and operational conditions for each stage, which are detailed in [Supporting Information Table S-2](#). Centrifugal pump models were used for all fluid pumping, with natural gas as the primary heating source and water for cooling. Equipment such as pumps, filters, mixers, and mills were powered by electrical motors.²⁴ The resulting beer stream contained 6% ethanol, water, residual sugar, humulene, and myrcene.

Life Cycle Assessment. In this study, LCA methodology was applied to evaluate the environmental performance of craft beer production, following the guidelines set by ISO 14040 and ISO 14044 standards.¹⁰ This approach allowed for a detailed analysis of the environmental impacts associated with each stage of the beer production process, considering the involved raw materials and utilities.

Goal and Scope. The main goals of this study are to quantify the overall environmental impacts associated with craft beer production, identify critical areas for improvement, and promote sustainable practices within the craft brewing industry. The functional unit was defined as 1 L of craft beer produced, and a “cradle-to-gate” approach was applied. This analysis excludes environmental impacts from industrial process equipment and pipeline fabrication due to data limitations, and does not include the stages of bottling, use, or disposal. This targeted assessment aims to inform brewers, policy-makers, and stakeholders about the most effective strategies for minimizing environmental impacts and fostering a more sustainable future for the craft beer industry.

Life Cycle Inventory. The life cycle inventory data was primarily collected from Aspen Plus® software simulations. Foreground data, estimated based on the defined functional unit of 1 L of craft beer, is presented in [Table 1](#). Data for malt, yeast, and hop production were adapted from existing literature, with inventories reported in [Supporting Information Tables S-3–S-5](#). All these inventories were modeled using OpenLCA 1.11 software, incorporating the Ecoinvent 3.8 database for the background system. This study implemented the Allocation at the Point of Substitution (APOS) system model, which proportionally allocates environmental burdens to specific processes based on their contributions.²¹ Transportation of raw materials, mainly imported from Europe, was a significant consideration in the analysis, with reference truck data provided in [Supporting Information Table S-6](#).

Life Cycle Impact Assessment. The environmental impacts of craft beer production were determined using the ReCiPe 2016 Midpoint (H) method.²² The impact categories considered in this study were fine particulate matter formation (FPMF) (kg PM_{2.5} equiv), fossil resource scarcity (FRS) (kg oil equiv), freshwater ecotoxicity (FE) (kg 1,4-DCB), freshwater eutrophication (FEut) (kg P equiv), global warming (GW) (kg CO₂ equiv), human carcinogenic toxicity (HCT) (kg 1,4-DCB), human noncarcinogenic toxicity (HNCT) (kg 1,4-DCB), ionizing radiation (IR) (kBq Co₆₀ equiv), land use (LU) (m² of a crop equiv), marine ecotoxicity (ME) (kg 1,4-DCB), marine eutrophication (MEP) (kg N equiv), mineral resource scarcity (MRS) (kg Cu equiv), ozone formation, human health (OFHH) (kg NO_x equiv), ozone formation, terrestrial ecosystems (OFTE) (kg NO_x equiv), stratospheric ozone depletion (SOD) (kg CFC11 equiv), terrestrial acidification (TA) (kg SO₂ equiv),

terrestrial ecotoxicity (TE) (kg 1,4-DCB), and water consumption (WC) (m³).

Uncertainty Analysis. To assess uncertainties in the impact results, a Monte Carlo simulation was conducted using OpenLCA software, consisting of 5000 iterations at a 95% confidence level. This statistical approach provided a robust basis for the uncertainty assessment. A pedigree matrix used to evaluate the quality and uncertainty of data sources for each raw material and utility included in the inventory. Log-normal distributions were employed to model the uncertainties for the Monte Carlo simulation.²³ Furthermore, the coefficient of variation (CV) was calculated for each impact category to quantify the variability in relation to the mean value. The CV, expressed as the percentage derived from the ratio of the standard deviation to the mean, provides a normalized measure of dispersion, facilitating comparison across different impact categories.²⁴

Economic Analysis. The economic analysis considered equipment dimensions and process input flows, including water, natural gas, electricity, and raw materials. The economic analysis considered Bogotá, Colombia, as the defined geographical location to produce craft beer. Detailed information on the required quantities and costs of raw materials for producing IPA beer is provided in [Supporting Information Table S-7](#).²⁵ Additionally, [Supporting Information Table S-8](#) presents the costs for rent and utilities such as electricity, natural gas, and tap water. This analysis aimed to assess the economic feasibility of craft beer production in terms of financial indicators over 10 years from 2023 to 2033.

This analysis evaluates the following economic indicators: weighted average cost of capital (WACC), net present value (NPV), modified internal rate of return (MIRR), benefit–cost ratio (BCR), and payback time. These indicators were assessed during the discounted cash flow method, commonly known as the present value model in economic engineering.²⁶ The WACC and payback time are critical indicators for potential investors, representing the minimum expected return and investment recovery period.²⁷ The NPV and MIRR provide a comprehensively understand of the economic returns and the investment profitability over the evaluated 10 year period. The MIRR is considered due to negative cash flows associated with capital investment at the project’s beginning and end.^{28,29} A BCR greater than 1.0 indicates a favorable economic outcome and is considered profitable.^{26,30} Detailed equations and descriptions of these economic indicators are reported in [Table S-9](#) in [Supporting Information](#).

The analysis also applied an inflation rate of 13.1% to annual production costs and selling beer prices, was set at 7000 Colombian Pesos (COP), equivalent to US\$ 1.4, during the first year of sales.^{31,32} This price aligns with the Colombian market’s standard rate for craft beer. Additionally, the market growth rate was set at 4.76%.^{33,34} The initial investment was primarily based on the equipment costs, including local materials, construction, and importation costs. These costs were calculated according to the dimensions obtained from the simulation and are outlined in [Supporting Information Table S-10](#).

Exergy Analysis. Exergy analysis is a critical step in evaluating the impacts associated with energy use in transformation processes, particularly in craft beer production.³⁵ This approach helps identify inefficiencies caused by irreversibilities in the process, and the information obtained from this analysis is highly valuable for identifying and enhancing the overall system’s efficiency and cost-effectiveness.³⁶

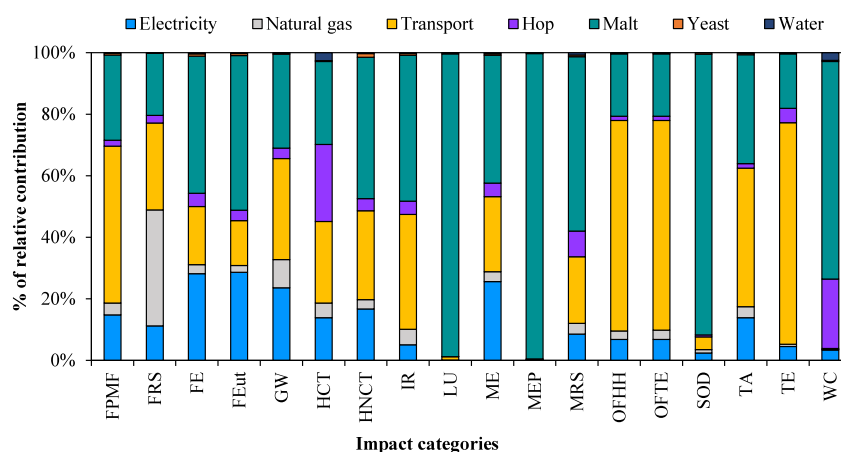


Figure 2. Contribution analysis of craft beer production, including electricity, natural gas, transport, and raw materials.

To quantify exergy destruction and exergetic efficiencies, Fuentes et al. proposed a phenomenologically based mathematical model that focuses on the exergetic balance of mass, heat, and work inputs and outputs.³⁷ This methodology accounts for the exergy destroyed due to irreversibilities at each stage of the brewing process. Physical exergy includes factors such as enthalpy, entropy, and dead state reference conditions, while chemical exergy refers to the inherent energy stored in chemical substances based on their composition and reactivity.^{38,39} The integration of both, physical and chemical transformations, highlight the suitability of this model for application to complex processes such as craft beer production.

In this study, the exergetic analysis was based on the equations reported in Fuentes et al. study to identify the specific stages where exergy is destroyed and determine opportunities for process improvement.³⁷ Key parameters, including flow rates, temperature, pressure, and composition, were determined under standard dead state conditions (1 atm and 293.15 K). Additionally, standard chemical exergy was calculated from semi-empiric correlations based on their elemental composition for molecules with complex chemical structures, such as cellulose and other biomass. Equation 1 was used to estimate a biomass compound's standard molar chemical exergy in kcal/kg.^{40,41}

$$\begin{aligned} \bar{e}_{0k}^{CH} = & 8177.79(C) + 27892.63(H) + 3173.66(O) \\ & + 0.15(O)[7837.667(C) + 33888.889(H) - 4236.1(O)] \end{aligned} \quad (1)$$

C, H, and O represent the mass fractions of carbon, hydrogen, and oxygen elements without halogens and sulfur.

Exergetic efficiencies were calculated and analyzed individually for each piece of equipment and globally across the entire process to identify the specific stages where improvement alternatives should be implemented.

RESULTS AND DISCUSSION

Contribution Analysis. Figure 2 presents the hotspot analysis for each impact category associated with the craft beer production process inputs. The impact assessment results for yeast, malt, and hops production are detailed in Supporting Information Table S-11.

The results in this study reveal that raw materials, particularly malt, had the highest impact across most impact categories. Specifically, malt was the most significant contributor in the categories of land use (LU) (0.38 m² of a crop equiv), marine eutrophication (MEP) (5.5 × 10⁻⁴ kg N equiv), stratospheric ozone depletion (SOD) (1.62 × 10⁻⁶ kg CFC11 equiv), and water consumption (WC) (3.77 × 10⁻² m³), with contributions exceeding 90%. These impacts are

primarily generated during malt production, with barley cultivation being a major hotspot, contributing over 95% to all impact categories. The agricultural practices involved in barley cultivation, including using fertilizers, pesticides, and extensive water usage, contribute substantially to these environmental impacts.^{42,43} Therefore, adopting sustainable farming practices, such as reduced fertilizer use, efficient irrigation systems, and integrated pest management, could decrease the environmental burden of barley cultivation.⁴⁴ Furthermore, sourcing barley from regions with more environmentally friendly agricultural practices could also help mitigate these impacts.⁴⁵

Another significant hotspot identified was the transportation required for importing raw materials, which notably impacted categories such as fine particulate matter formation (FPMF) (4.12 × 10⁻⁴ kg PM_{2.5} equiv), ozone formation (OFHH and OFTE) (1.3 × 10⁻³ kg NO_x equiv), and terrestrial ecotoxicity (TE) (1.76 kg 1,4-DCB). These impacts are primarily due to emissions from fossil fuel combustion in ships and trucks importing raw materials. These emissions include particulate matter, nitrogen oxides (NO_x), methane, and volatile organic compounds (VOCs), which are released into the atmosphere.^{46,47}

Overall, the analysis shows that raw materials accounted approximately for 51% of craft beer production's total life cycle impact for most impact categories. Transport contributed approximately 31%, while electricity and natural gas accounted for 13% and 5% of the impacts, respectively. These findings highlight the importance of focusing on raw material sourcing and transportation challenges to mitigate the environmental footprint of craft beer production.

In the global warming impact category, our study estimated greenhouse gas emissions at 0.39 kg CO₂ equiv/(L of beer produced). This value is notably lower than those reported in similar LCA studies conducted in microbreweries in both Brazil and Wales, United Kingdom. For instance, a Brazilian microbrewery study reported emissions of 2.46 kg CO₂ equiv/L, while research on seven microbreweries in Wales found values ranging from 0.76 to 1.90 kg CO₂-eq/L.^{48,49} The higher emissions observed in these studies can be attributed to factors such as the smaller scale of operations in microbreweries as well as the inclusion of the bottling and packaging stages in their assessments.

Previous research has shown that these stages significantly influence the overall carbon footprint, contributing an

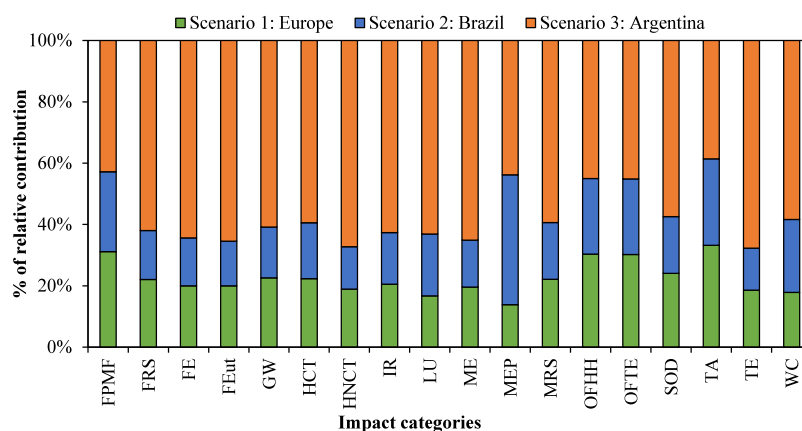


Figure 3. Sensitivity analysis of transportation for raw material sourcing in craft beer production.

estimated 50% of total CO₂ equiv emissions.⁵⁰ However, our study did not assess the bottling and packaging stages due to the absence of primary data on the specific packaging practices of the brewery analyzed. We therefore recommend that future studies incorporate the environmental impacts of packaging to achieve a more comprehensive life cycle analysis of brewing systems.

To further reduce emissions, solar photovoltaic (PV) technologies have been explored for energy-intensive brewing stages. A study by Diniz et al. demonstrated that substituting 20% of grid electricity with solar PV in a Brazilian microbrewery can avoid approximately 175 kg CO₂ equiv/year.⁵¹ The progressive adoption of renewable energy sources like solar PV represents a viable strategy to enhance the environmental performance of craft beer production.

Sensitivity Analysis for Transport. The environmental hotspot analysis revealed that transport is a major contributor to most impact categories in craft beer production. This sensitivity analysis evaluates the environmental impacts related to the transportation of raw materials for craft beer production in Bogotá, Colombia, by comparing three sourcing scenarios: Europe, Brazil, and Argentina. These regions were selected based on their relevance to Latin America breweries and the ability to supply high-quality brewing ingredients that maintain the beer's sensory profiles.^{52,53} Europe is a traditional supplier with advanced agricultural practices and a long history of brewing, while Brazil and Argentina are major agricultural producers in Latin America with significant barley and malt production.^{54–58}

Figure 3 shows the environmental performance of the three transport scenarios considered in this study. In general, scenario 1 (Europe) and scenario 2 (Brazil) demonstrated notably lower environmental impacts. For example, in terms of Fossil Resource Scarcity (FRS), scenario 3 (Argentina) exhibited the highest impact (0.12 kg oil eq), whereas scenario 1 (Europe) (0.042 kg oil eq) and scenario 2 (Brazil) (0.03 kg oil eq), demonstrated significantly lower impacts, 641% and 74% reductions, respectively, compared to Argentina.

Overall, the results indicated that scenario 2 (Brazil) was the most environmentally favorable option, reducing impacts by approximately 22% compared to scenario 1 (Europe), due to the geographical proximity to Colombia. Although scenario 1 (Europe) offers some potential for reducing impacts, particularly through optimizing transport routes, it does not match the benefits of the Brazilian scenario. Moreover, scenario 3 (Argentina) exhibited the highest environmental

impacts across all categories, approximately 58% of contribution. These results were driven by less efficient transport infrastructure, long road transport distances, and multiple transport modes, leading to increased fossil fuel consumption and emissions.

This study highlights the importance of transportation in the environmental footprint of craft beer production and demonstrates that sourcing raw materials from Brazil (scenario 2) offers the best environmental performance compared to European and Argentine alternatives. These findings underscore the importance of regional sourcing strategies in minimizing environmental burdens while ensuring product quality.

To provide an integrated perspective on sourcing decisions, we evaluated the economic implications of the three transportation scenarios alongside their environmental impacts. Sourcing raw materials from Brazil (scenario 2) not only reduced environmental impacts by approximately 22% compared to European sourcing (scenario 1) but also resulted in an estimated 18% reduction in transportation costs. This cost difference is primarily attributed to reduced shipping distances and fewer transshipment requirements. Conversely, the Argentinian scenario (scenario 3), despite its geographical proximity to Colombia, incurred higher costs due to less efficient logistics infrastructure and multiple transport mode changes. These findings demonstrate how environmentally preferable sourcing options can often align with economic advantages, though the specific cost–benefit ratio will depend on commodity prices, shipping rates, and potential quality differences between regional suppliers.

Uncertainty Analysis. Uncertainty results are summarized in Supporting Information Table S-12, which provides detailed information for each impact category. This includes mean values, standard deviations, and the 5th and 95th percentiles, offering a comprehensive overview of the distribution and range of the impact results. Additionally, the coefficients of variation (CV) were calculated to assess the relative dispersion of impact estimates. The CV percentages below 12% indicate a relatively low degree of variability, suggesting a high level of confidence in the results. To enhance the interpretation of uncertainty distributions, boxplots figures were generated and are included in Figure S-2 of the Supporting Information.

Comparing the results with previous studies, carbon footprint estimates for beer have been reported in the range of 0.8 and 1.3 kg CO₂ equiv/L, with a 95% confidence interval.^{59,60} In contrast, the results of our Monte Carlo

Table 2. Economic Indicators of Selected Beverage Production Projects Across Different Locations

	Craft beer ⁶¹	Wine ⁶²	Beer ⁶³	Beer ⁶⁴	Soft drinks ⁶⁵	Pineapple wine ⁶⁶
Location	Indiana (USA)	Georgia	Chile	Spain	Brazil	Colombia
Annual production	480000 L	750000 L	1375000 L	93600 L	6582.5 tons	1600 tons
Period (years)	7	10	5	10	10	27
Payback time (years)	13	6	7	4,3	0,8	
NPV (US\$)	−48405	2263	84365		2510,825	4,776
IRR (%)	−15.09	20.10	18.00		49.77	23

simulations indicate a lower range, between 0.32 and 0.47 kg CO₂ equiv/L. This finding confirms that our assessment is not only more environmentally favorable but also methodologically stronger, providing a more accurate and reliable environmental footprint of craft beer production.

Economic Analysis. The economic analysis applied an inflation rate of 13.1% to annual production costs and selling beer prices, based on official Colombian economic data during the study period.^{31,32} While this rate may appear high compared to developed economies, it represents the actual market conditions facing craft brewers in Colombia and reflects regional economic realities. This approach ensures that our economic projections account for the specific challenges of operating a craft brewery in this emerging market.

These economic findings complement our environmental and exergetic analyses by identifying financial motivations that align with sustainability improvements. For example, the exergy analysis identified filtration as responsible for 68% of thermodynamic losses. Investments in more efficient filtration technology, while requiring initial capital expenditure, would simultaneously reduce resource waste (improving environmental performance) and enhance extraction efficiency (improving economic returns). Similarly, our environmental analysis highlighted raw material transport as a significant impact contributor (31%), which directly connects to procurement costs in the economic model. By optimizing sourcing strategies, as explored in our sensitivity analysis, brewers can simultaneously reduce environmental impacts and operating expenses.

The economic analysis presented in this study underscores the strong financial competitiveness of craft beer production, exceeding the minimum profitability threshold for investors, see [Supporting Information Table S-13](#). With a MIRR of 16.92% and a BCR of 3.02, the results support a high return on investment, generating US\$ 3.02 for every dollar invested.

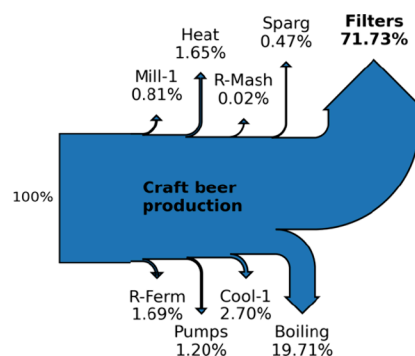
A comparative economic analysis was conducted, including various beverage sectors, including soft drinks, wine, and traditional beer.^{61–66} [Table 2](#) summarizes the key economic indicators considered, such as annual production, payback time, NPV, IRR, and overall financial viability of each production process. This comparative approach highlights the economic differences between these beverage types, providing valuable context for the economic feasibility of the proposed craft beer production.

Results for craft beer production in Indiana indicated negative a NPV and IRR with a 13 year payback period, which reflects its financial unfeasibility. Similarly, wine production in Georgia showed a low NPV and only marginal financial gains. In contrast, beer production in Chile reported a higher NPV; however, the 7 year payback period may discourage investors who prioritize quicker returns. The microbrewery in Spain presented a 4.3 year payback time, which, despite being more favorable, could be considered lengthy for potential investors.

Other industries, such as soft drinks production in Brazil, demonstrated exceptionally high IRR and NPV; however, these results stem from large-scale operations that differs from the focus of our study. Conversely, pineapple wine production in Colombia yielded a relatively low NPV, indicating limited absolute value generation.

These findings support our proposed craft beer production, showcasing robust financial outcomes in terms of both value creation and investment safety. Overall, the analysis demonstrates that this study not only offers a highly attractive and rapid financial return but also stands out as a safer and more profitable investment opportunity when compared to other projects that exhibit greater uncertainties, longer recovery times, or potential financial losses.

Exergy Analysis. The exergy analysis, conducted through process simulation in Aspen Plus, provided detailed insights into the energy efficiency of craft beer production. A Sankey diagram was employed to analyze each process equipment's exergy destruction and distribution ([Figure 4](#)). The findings

**Figure 4.** Sankey diagram for exergy analysis. This diagram depicts the distribution of exergy across various stages of the brewing process.

revealed that the largest exergy destruction, 68% of total losses, occurred during the first filtration (FILT-1) in the wort separation stage. In this process the spent grain (represented by amylose, cellulose, and glucose) is discarded as waste. These streams have a high standard molar chemical exergy, meaning they possess considerable exergy potential that is being wasted. The subsequent filtration stages (FILT-2 and FILT-3) exhibited lower exergy destruction, as spent hop and yeast streams had lower flow rates and chemical exergy. The boiling stage was the second major contributor, with 19.7% of exergy destruction, mainly due to high heat demand and water evaporation. Other units, such as mills, pumps, and heat exchangers, exhibited minimal inefficiencies, accounting for less than 2% of total exergy losses. Moreover, during the maceration process (R-MASH), only the enzymatic conversion of amylose was considered, and no waste was generated. These results stem from the mechanical inefficiencies of the equipment, which exhibit low dissipation and generate no

significant waste streams, thereby contributing minimally to overall exergy destruction. Other processes, such as sparging (SPARG), cooling (COOL-1), and fermentation (R-FERM), demonstrated lower notable contributions to overall exergy destruction, primarily due to heat dissipation and CO₂ emissions.

A more detailed analysis was conducted to assess energy utilization throughout the craft beer process, examining the specific exergetic transformations and efficiencies. Results are summarized in [Supporting Information](#) Table S-14. This analysis revealed that total input exergy—including substance flows, heat, and work—exceeded output exergy due to irreversible losses within the system. These losses are categorized as internal irreversibilities ($\dot{E}_{Di}$), which result from thermodynamic inefficiencies, and external irreversibilities ($\dot{E}_{Dw}$), which arise from waste streams and heat dissipation.⁶⁷ Both contribute to overall system exergy destruction ($\dot{E}_{DT}$), with significant implications for energy efficiency.

Findings showed that mass flows contributed 86.1% of the total exergy input, with barley malt as the major contributor due to its high chemical exergy. In contrast, work exergy was minimal (3.3%), given the low power consumption of electric motors operating pumps and mills. Boiling accounted for 84.2% of total heat exergy transfer, reinforcing its role as a key target for efficiency improvements.

Regarding exergy destruction, internal irreversibilities contributed 27% of total losses, while waste streams, including spent grains, spent hops, CO₂, and steam, accounted for 73%. The global exergy efficiency was 54.3%, meaning that nearly half of the exergy input is lost. However, when considering the potential reuse of waste streams, the rational efficiency reached 87.7%, suggesting an improvement potential of 33.4% through waste minimization and energy recovery strategies.

To enhance energy efficiency, strategies should focus on optimizing the mashing stage. Previous research suggests that combining decoction and infusion techniques can improve starch-to-sugar conversion while reducing energy consumption.⁶⁸ Adjusting dilution ratios (e.g., increasing malt concentration) and incorporating a filter press could also enhance sugar recovery post-sparging. Moreover, optimizing the water-to-malt ratio to enhance extraction efficiency, or integrating a filter press to recover additional sugars, could reduce energy losses.⁶⁹

Ultimately, this study highlights the importance of integrating exergy, environmental, and economic analyses to optimize craft beer production. The synthesis of these results provides a strategic framework for breweries, offering clear insights into how improvements in energy efficiency, cost management, and environmental impact reduction can be achieved simultaneously.

■ LIMITATIONS AND FUTURE WORK

This study presents several limitations that offer opportunities for future research. First, our analysis excludes the packaging stage, which has been identified in other studies as a significant contributor to the overall environmental impact of beer production. Future research should integrate production and packaging analyses for a more comprehensive assessment.

Second, while our findings highlight the environmental impact of agricultural practices in raw material production, we recognize that craft brewers typically have limited direct

influence over these practices. However, the information provided can guide brewers in making more informed sourcing decisions and potentially advocate for improvements within their supply chains.

Third, our three analytical approaches (environmental, economic, and exergetic), while presented sequentially for clarity, are interconnected and provide complementary insights. For example, the exergy destruction identified in the filtration process (68%) directly relates to both economic costs through wasted resources and environmental impacts through raw material consumption. Future work could develop integrated sustainability metrics that simultaneously capture these dimensions.

Finally, our analysis focused on a specific craft beer style (IPA) in a Colombian context. Results may vary for different beer styles or regional conditions, suggesting the need for comparative studies across various brewing contexts. Additionally, the valorization of brewing by-products, particularly spent grains, represents an important area for future research that could further improve both economic and environmental performance.

■ CONCLUSIONS

This study assesses the sustainability of craft beer production, integrating environmental, economic, and exergetic aspects. Aspen Plus® software was used for simulations, determining the required mass flows and equipment dimensions. An LCA was performed by employing inventory data from the proposed simulation process, accounting for the uncertainties related to the input data. Furthermore, this study provides open-source LCI datasets for each estimated raw material in craft beer production.

The environmental analysis identified malt production and transport as key hotspots, with contributions exceeding 90% and 70%, respectively, in some impact categories. The environmental impacts associated with malt production primarily arise from barley cultivation, which contributed over 95% to all impact categories. In this process, agricultural practices like fertilizers, pesticides, and extensive water consumption play a significant role. Additionally, this study highlighted the substantial environmental burden of transport, and a sensitivity analysis was carried out with different scenarios for raw materials sourcing. Results showed that sourcing raw materials from Brazil have a favorable environmental performance, reducing impacts by approximately 22% compared to scenario 1 (Europe).

Economically, results revealed that this project offers a rapid financial return of US\$ 3.02 for every dollar invested. Moreover, it stands out as a safer and more profitable investment opportunity compared to others with higher uncertainties, longer recovery times, or potential economic losses. Moreover, these findings highlight the capacity of our proposed craft beer production to deliver substantial returns in a competitive market.

Exergetic analysis revealed that 68% of exergy loss occurs during the filtration in the wort separation stage, suggesting potential for energy savings and reduced processing time. These improvements suggest a potential area for future research and development in craft beer production.

This study provides valuable guidance for decision-making processes and proposes fundamental advances for promoting sustainable industrial practices in craft beer production. This study demonstrated that by adopting sustainable sourcing

practices, such as selecting raw materials from regions with more environmentally friendly agricultural practices, breweries can reduce their environmental impacts and align with growing consumer demand for sustainable products. Additionally, by implementing energy-saving strategies, breweries can reduce energy consumption, lower costs, and improve profitability, reinforcing the industry's economic competitiveness while aligning with environmental goals. Moreover, there is a clear need for continued research into new or improved processing technologies, such as energy-efficient brewing methods, waste valorization techniques, and resource-saving innovations.

This comprehensive assessment provides craft brewers with practical insights for immediate implementation while highlighting areas for further innovation. By integrating environmental, economic, and exergetic analyses, we have identified key intervention points—particularly in raw material sourcing, transportation logistics, and filtration efficiency—that offer opportunities for simultaneous improvement across multiple sustainability dimensions.

The research makes a significant contribution by providing open-source LCI datasets for craft beer production inputs and demonstrating the feasibility of applying integrated sustainability assessment approaches to small-scale brewing operations. Future research should expand this foundation by incorporating packaging analysis, exploring waste valorization opportunities, and investigating how renewable energy integration might address the inefficiencies identified in our exergy analysis.

Ultimately, the craft beer sector's sustainable development will depend on holistic approaches that recognize the interconnections between environmental impact, economic viability, and resource efficiency—a framework this study has established for future investigations and practical applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssusresmgmt.4c00374>.

Aspen Plus flowsheet; list of unit operation blocks considered in Aspen Plus; process modeling of stages; inventories of malt production adapted from Kløverpris et al. and of yeast production adapted from Møller et al.; hop productin inventory; distances, raw material costs, and utility costs considered; economic analysis equations; craft microbrewery equipment costs; impact assessment results of yeast, malt, and hop production; uncertainties results for assessed impact categories; impact category boxplots; economic analysis results; exergy flows and global exergy efficiencies; list of abbreviations (PDF)

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■ REFERENCES

- (1) Garavaglia, C.; Swinnen, J. *Economic Perspectives on Craft Beer*; Garavaglia, C., Swinnen, J., Eds.; Springer International: Cham, Switzerland, 2018. DOI: [10.1007/978-3-319-58235-1](https://doi.org/10.1007/978-3-319-58235-1).
- (2) Baiano, A. Craft Beer: An Overview. *Compr Rev. Food Sci. Food Saf* **2021**, 20 (2), 1829–1856.
- (3) Gobbi, L.; Stanković, M.; Ruggeri, M.; Savastano, M. Craft Beer in Food Science: A Review and Conceptual Framework. *Beverages* **2024**, 10 (3), 91.
- (4) Toro-Gonzalez, D. The Craft Brewing Industry in Latin America: The Case of Colombia. *Economic Perspectives on Craft Beer: A Revolution in the Global Beer Industry* **2018**, 115–136.
- (5) Latin America Craft Beer Market Size (2024 to 2032). *Market Data Forecast*; 2024. <https://www.marketdataforecast.com/market-reports/la-craft-beer-market> (accessed 2025-01-24).
- (6) BarthHaas. *BarthHaas Report 2021/2022, 2023*. https://www.barthhaas.com/fileadmin/user_upload/01-barthhaas-2022/Downloads/Reports_Broschures/BarthHaas_Reports/English/2020-2030/BarthHaas_Report_2021_2022.pdf (accessed 2023-07-08).
- (7) Jones, E. Brewing Green: Sustainability in the Craft Beer Movement. *Craft Beverages and Tourism, Vol. 2: Environmental, Societal, and Marketing Implications*; Slocum, S. L., Kline, C., Cavaliere, C. T., Eds.; Springer International: Cham, Switzerland, 2018; pp 9–26. DOI: [10.1007/978-3-319-57189-8_2](https://doi.org/10.1007/978-3-319-57189-8_2).
- (8) Hoalst-Pullen, N.; Patterson, M. W.; Mattord, R. A.; Vest, M. D. Sustainability Trends in the Regional Craft Beer Industry. *The Geography of Beer: Regions, Environment, and Societies*; Patterson, M., Hoalst-Pullen, N., Eds.; Springer: Dordrecht, The Netherlands, 2014; pp 109–116. DOI: [10.1007/978-94-007-7787-3_11](https://doi.org/10.1007/978-94-007-7787-3_11).
- (9) Rahman, I.; Nanu, L.; Sozen, E. The Adoption of Environmental Practices in Craft Breweries: The Role of Owner-Managers' Consumption Values, Motivation, and Perceived Business Challenges. *J. Clean Prod* **2023**, 416, 137948.

- (10) ISO. ISO 14040:2006 *Environmental management—Life cycle assessment—Principles and framework*, 2006. <https://www.iso.org/standard/37456.html> (accessed 2025-03-20).
- (11) Roy, P.; Nei, D.; Orikasa, T.; Xu, Q.; Okadome, H.; Nakamura, N.; Shiina, T. A Review of Life Cycle Assessment (LCA) on Some Food Products. *J. Food Eng.* **2009**, *90* (1), 1–10.
- (12) Garavaglia, C.; Swinnen, J. Economics of the Craft Beer Revolution: A Comparative International Perspective. *Economic Perspectives on Craft Beer: A Revolution in the Global Beer Industry*; Garavaglia, C., Swinnen, J., Eds.; Springer International: Cham, Switzerland, 2018; pp 3–51. DOI: 10.1007/978-3-319-58235-1_1.
- (13) Goldammer, T. *The Brewer's Handbook: The Complete Book to Brewing Beer*; Apex, 2008.
- (14) Worrell, E.; Galitsky, C.; Martin, N. *Energy Efficiency Opportunities in the Brewery Industry*; Lawrence Berkeley National Laboratory, 2002. DOI: 10.1712/881595.
- (15) Mosher, M.; Trantham, K. Overview of the Brewing Process. *Brewing Science: A Multidisciplinary Approach*; Springer International: Cham, Switzerland, 2021; pp 103–131. DOI: 10.1007/978-3-030-73419-0_4.
- (16) Jana, A. K. *Chemical Process Modelling and Computer Simulation*, 3rd ed.; PHI Learning, 2018.
- (17) Haydary, J. *Chemical Process Design and Simulation: Aspen Plus and Aspen Hysys Applications*; Wiley, 2018. DOI: 10.1002/9781119311478.
- (18) Zarate, C.; Cruz, J. C.; Reyes, L. H. Performance of Craft Beer Local Breweries in Bogota, Colombia: Sustainable Engineering View.
- (19) Usosky, D.; Kim, K.; Moon, J.-H.; Jung, K.-D.; Lee, U. Isobaric Vapor–Liquid Equilibria for the Formic Acid–N-Methyl-2-Pyrrolidone Binary System at 50, 20, and 10 KPa and Modeling Using the NRTL-HOC and PC-SAFT Models. *J. Chem. Eng. Data* **2021**, *66* (12), 4516–4525.
- (20) Caudle, B.; Vieira, M. C.; Gorensek, M. B.; Chen, C.-C. Modeling Phase Equilibrium of Common Sugars Glucose, Fructose, and Sucrose in Mixed Solvents. *J. Chem. Eng. Data* **2021**, *66* (11), 4193–4205.
- (21) Pavlović, A.; Donchev, T.; Petkova, D.; Staletović, N. Sustainability of Alternative Reinforcement for Concrete Structures: Life Cycle Assessment of Basalt FRP Bars. *Constr. Build. Mater.* **2022**, *334*, 127424.
- (22) Huijbregts, M. A. J.; Steinmann, Z. J. N.; Elshout, P. M. F.; Stam, G.; Veronesi, F.; Vieira, M.; Zijp, M.; Hollander, A.; van Zelm, R. ReCiPe2016: A Harmonised Life Cycle Impact Assessment Method at Midpoint and Endpoint Level. *International Journal of Life Cycle Assessment* **2017**, *22* (2), 138–147.
- (23) Sonnemann, G. W.; Schuhmacher, M.; Castells, F. Uncertainty Assessment by a Monte Carlo Simulation in a Life Cycle Inventory of Electricity Produced by a Waste Incinerator. *J. Clean Prod* **2003**, *11* (3), 279–292.
- (24) Guo, M.; Murphy, R. J. LCA Data Quality: Sensitivity and Uncertainty Analysis. *Science of the Total Environment* **2012**, *435*–436, 230–243.
- (25) Barczy, A.; Szabó, D.; Magyari, P.; Gécsi, G. Colour Analysing of IPA Brewing Phases, Including Daily Measuring of the Fermentation Process. *Science, Technology and Innovation* **2022**, *15* (1), 1–7.
- (26) Sullivan, W.; Wicks, E.; Luxhoj, J. *Ingeniería Económica de DeGarmo*, 12th ed.; Pearson Education, 2004.
- (27) Agnes Cheng, C. S.; Kite, D.; Radtke, R. THE APPLICABILITY AND USAGE OF NPV AND IRR CAPITAL BUDGETING TECHNIQUES. *Managerial Finance* **1994**, *20* (7), 10–36.
- (28) Range, A. d. S.; Santos, J. C. de S.; Savoia, J. R. F. Modified Profitability Index and Internal Rate of Return. *Journal of International Business and Economics* **2016**, *4* (2), DOI: 10.15640/jibe.v4n2a2.
- (29) Iankovyi, V. Comparative Analysis of the Most Important Economic Criteria for the Company's Investment Projects. *Socio-Economic Research Bulletin* **2021**, *2* (77), 125–141.
- (30) Anaman, K. A.; Atzeni, M. G.; Mayer, D. G.; Stuart, M. A. Benefit-Cost Analysis of the Use of Sterile Insect Technique to Eradicate Screwworm Fly in the Event of an Invasion of Australia. *Prev. Vet. Med.* **1994**, *20* (1–2), 79–98.
- (31) DANE. DANE—IPC Información Técnica. <https://www.dane.gov.co/index.php/estadisticas-por-tema/precios-y-costos/indice-de-precios-al-consumidor-ipc/ipc-informacion-tecnica> (accessed 2024-07-10).
- (32) Portafolio. Colombia tiene la cerveza más barata entre países de América Latina: cuánto cuesta? <https://www.portafolio.co/internacional/cuanto-cuesta-en-promedio-conseguir-una-cerveza-en-latinoamerica-610309> (accessed 2024-12-15).
- (33) Mercado de la cerveza—Compartir, tendencias y análisis de la industria. <https://www.mordorintelligence.com/es/industry-reports/beer-market> (accessed 2024-07-09).
- (34) Pradeep, N. Craft Beer Market Research Report; 2024. <https://www.marketresearchfuture.com/reports/craft-beer-market-973> (accessed 2024-07-10).
- (35) Jemigbeyi, O. S.; Salau, T. A. O.; Oyewola, O. M. Energetic, Exergetic, and Exergoeconomic Analyses of Beer Wort Production Processes. *Front. Energy Res.* **2024**, *12*, 1456688.
- (36) Tsatsaronis, G.; Czesla, F. Thermoeconomics. *Encyclopedia of Physical Science and Technology*, 3rd ed.; Meyers, R. A., Ed.; Academic Press: New York, 2003; pp 659–680. DOI: 10.1016/B0-12-227410-5/00944-3.
- (37) Fuentes, O. P.; Trujillo, D. M.; Sánchez, M. E.; Abrego-Perez, A. L.; Osma, J. F.; Cruz, J. C. Embracing Sustainability in the Industry: A Study of Environmental, Economic, and Exergetic Performances in Large-Scale Production of Magnetite Nanoparticles. *ACS Sustain. Chem. Eng.* **2024**, *12* (2), 760–772.
- (38) Morosuk, T.; Tsatsaronis, G. Splitting Physical Exergy: Theory and Application. *Energy* **2019**, *167*, 698–707.
- (39) Michalakakis, C.; Fouillou, J.; Lupton, R. C.; Gonzalez Hernandez, A.; Cullen, J. M. Calculating the Chemical Exergy of Materials. *J. Ind. Ecol.* **2021**, *25* (2), 274–287.
- (40) Shieh, J. H.; Fan, L. T. Estimation of Energy (Enthalpy) and Exergy (Availability) Contents in Structurally Complicated Materials. *Energy Sources* **1982**, *6* (1–2), 1–46.
- (41) Vasquez, C. A.; Marulanda, M.; Sánchez, M.; Cruz, J. C. SIMULATION AND SUSTAINABLE ENGINEERING ANALYSIS OF CELLULOSE PRODUCTION VIA IONIC LIQUIDS. *Rev. Mex. Ing. Quim.* **2019**, *18* (3), 1037–1050.
- (42) Bessou, C.; Basset-Mens, C.; Tran, T.; Benoist, A. LCA Applied to Perennial Cropping Systems: A Review Focused on the Farm Stage. *Int. J. Life Cycle Assess.* **2013**, *18* (2), 340–361.
- (43) Tilman, D. Global Environmental Impacts of Agricultural Expansion: The Need for Sustainable and Efficient Practices. *Proc. Natl. Acad. Sci. U. S. A.* **1999**, *96* (11), 5995–6000.
- (44) Tricase, C.; Lamonaca, E.; Ingrao, C.; Bacenetti, J.; Lo Giudice, A. A Comparative Life Cycle Assessment between Organic and Conventional Barley Cultivation for Sustainable Agriculture Pathways. *J. Clean Prod.* **2018**, *172*, 3747–3759.
- (45) Stylianou, M.; Papamichael, I.; Voukaki, I.; Tsangas, M.; Omirou, M.; Ioannides, I. M.; Zorpas, A. A. LCA of Barley Production: A Case Study from Cyprus. *Int. J. Environ. Res. Public Health* **2023**, *20* (3), 2417.
- (46) Jørgensen, A.-M. M.; Ywema, P. E.; Frees, N.; Exner, S.; Bracke, R. Transportation in LCA. *Int. J. Life Cycle Assess.* **1996**, *1* (4), 218–220.
- (47) Omahne, V.; Krajnc, D.; Kovačič Lukman, R. A Critical Overview of Scientific Publications on Life Cycle Assessment in Transport-Related Topics. *Clean Technol. Environ. Policy* **2021**, *23* (3), 711–730.
- (48) Diniz, D. d. P.; Carvalho, M. Environmental Repercussions of Craft Beer Production in Northeast Brazil. *Sustainability* **2024**, *16* (11), 4566.
- (49) Morgan, D. R.; Styles, D.; Lane, E. T. Thirsty Work: Assessing the Environmental Footprint of Craft Beer. *Sustain. Prod. Consum.* **2021**, *27*, 242–253.

- (50) Pinto da Silva, L.; Esteves da Silva, J. C. G. Evaluation of the Carbon Footprint of the Life Cycle of Wine Production: A Review. *Cleaner and Circular Bioeconomy* **2022**, 2, 100021.
- (51) de Paula Diniz, D.; Carvalho, M.; Abrahao, R. Greenhouse Gas Accounting for the Energy Transition in a Brewery. *Environ. Prog. Sustainable Energy* **2021**, 40 (2), No. e13563.
- (52) da Rosa Almeida, A.; Maciel, M. V. d. O. B.; Cardoso Gasparini Gandolpho, B.; Machado, M. H.; Teixeira, G. L.; Bertoldi, F. C.; Noronha, C. M.; Vitali, L.; Block, J. M.; Barreto, P. L. M. Brazilian Grown Cascade Hop (*Humulus Lupulus* L.): LC-ESI-MS-MS and GC-MS Analysis of Chemical Composition and Antioxidant Activity of Extracts and Essential Oils. *Journal of the American Society of Brewing Chemists* **2021**, 79 (2), 156–166.
- (53) Jastrombek, J. M.; Faguerazzi, M. M.; de Cassio Pierezan, H.; Rufato, L.; Sato, A. J.; da Silva Ricce, W.; Marques, V. V.; Leles, N. R.; Roberto, S. R. An Emerging Crop in Subtropical Areas in Brazil. *Horticulturae* **2022**, 8 (5), 393.
- (54) Olajire, A. A. The Brewing Industry and Environmental Challenges. *J. Clean Prod* **2020**, 256, 102817.
- (55) Almeida, A. da R.; Maciel, M. V. de O. B.; Machado, M. H.; Bazzo, G. C.; de Armas, R. D.; Vitorino, V. B.; Vitali, L.; Block, J. M.; Barreto, P. L. M. Bioactive Compounds and Antioxidant Activities of Brazilian Hop (*Humulus Lupulus* L.) Extracts. *Int. J. Food Sci. Technol.* **2020**, 55 (1), 340–347.
- (56) Rodrigues Arruda, T.; Fontes Pinheiro, P.; Ibrahim Silva, P.; Campos Bernardes, P. Exclusive Raw Material for Beer Production? Addressing Greener Extraction Techniques, the Relevance, and Prospects of Hops (*Humulus Lupulus* L.) for the Food Industry. *Food Bioproc Tech* **2022**, 15 (2), 275–305.
- (57) Otero, E. A.; Miralles, D. J.; Peton, A.; Conti, V. A.; Giménez, F. J.; Benech-Arnold, R. L. On-Field Assessment of the Environmental Modulation of Malting Quality in Barley Crops. *Field Crops Res.* **2021**, 271, 108252.
- (58) Olivera Rodriguez, P.; Holzman, M. E.; Mujica, C. R.; Rivas, R. E.; Aldaya, M. M. Different Approaches to Assessing Pollution Load: The Case of Nitrogen-Related Grey Water Footprint of Barley and Soybean in Argentina. *Water (Basel)* **2021**, 13 (24), 3558.
- (59) D'Ascenzo, F.; Vinci, G.; Maddaloni, L.; Ruggeri, M.; Savastano, M. Application of Life Cycle Assessment in Beer Production: Systematic Review. *Beverages* **2024**, 10 (3), 86.
- (60) Zheng, H.; Xing, M.; Cao, T.; Zhang, J. Uncertainty Study and Parameter Optimization of Carbon Footprint Analysis for Fermentation Cylinder. *Sustainability* **2019**, 11 (3), 661.
- (61) Salazar T., M. B.; San Martin-Gonzalez, M. F.; Cai, H.; Huang, J.-Y. Economic and Environmental Performance of Instantaneous Water Heating System for Craft Beer Production. *Food and Bioproducts Processing* **2021**, 127, 472–481.
- (62) Chkareuli, V.; Darguashvili, G.; Atstaja, D.; Susniene, R. Assessing the Financial Viability and Sustainability of Circular Business Models in the Wine Industry: A Comparative Analysis to Traditional Linear Business Model—Case of Georgia. *Sustainability* **2024**, 16 (7), 2877.
- (63) Crespo, A.; Muñoz, C.; Cerda, M. T.; Muñoz, I.; Hernández, C.; Ramirez, G.; Ibarra, M.; Dinter, F. Techno-economic Analysis of Solar Photovoltaics and Solar Thermal Energy Integration in a Chilean Brewery. In *Proceedings of the ISES Solar World Congress 2019*; International Solar Energy Society: Freiburg, Germany, 2019; pp 1–11. DOI: 10.18086/swc.2019.12.02.
- (64) Pino, A.; Pino, F. J.; Guerra, J. Integration of Solar Energy in Small-Scale Industries: Application to Microbreweries. *Sustainable Energy Technologies and Assessments* **2023**, 57, 103276.
- (65) Cotrim, W. d. S.; Cotrim, K. C. F.; Coimbra, J. C. Modeling and analysis of economic scenarios of the whey-based carbonated beverage production process. *Journal of Bioenergy and Food Science* **2019**, 6 (4), 97–108.
- (66) Palacios, J. F. M.; Barrera, R.; Zondervan, E. Process Design and Techno-Economic Analysis of a Pineapple Wine Production Plant under the Context of the Choco-Colombia Region. *Computer Aided Chemical Engineering* **2019**, 46, 277–282.
- (67) Sorin, M.; Lambert, J.; Paris, J. Exergy Flows Analysis in Chemical Reactors. *Chemical Engineering Research and Design* **1998**, 76 (3), 389–395.
- (68) Montanari, L.; Floridi, S.; Marconi, O.; Tironzelli, M.; Fantozzi, P. Effect of Mashing Procedures on Brewing. *European Food Research and Technology* **2005**, 221 (1–2), 175–179.
- (69) De Schepper, C. F.; Courtin, C. M. High Mashing Thickness Negatively Influences Gelatinisation of Small and Large Starch Granules and Starch Conversion Efficiency during Barley Malt Brewing. *Food Hydrocoll* **2022**, 131, 107745.